From a moving restriction to merge receiving

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Scope. An isolated source-free receiving subproblem with a *prescribed* zero-passing path, followed by conditional node/accounting checks. No endogenous metered scenario, complete joined solver, empirical calibration or useful gradient is certified. Future E1 births and lateral sources are absent from the diagnostic only; they are not removed from the admitted task. Full-source provenance is in `joined_receiving_source_audit.md`.

1. The state at a moving restriction is not the state at the merge

Let $x = 0$ be the receiving-road entrance. Use a triangular fundamental diagram with free speed $u > 0$, backward-wave magnitude $w > 0$ and jam density $\kappa > 0$:

$$F(\rho) = \min\{u\rho, w(\kappa - \rho)\}, \qquad Q = \frac{uw\kappa}{u + w}. \tag{1}$$

Units: u, w, v in m/s; ρ, κ in vehicle equivalents/m; F, Q in vehicle equivalents/s. Start with an empty receiving road and one compatible boundary-born prescribed restriction $\phi(0) = 0$, $\dot{\phi} = v(\tau) \in [0, u]$. The downstream boundary is outside the diagnostic's domain of dependence. Let

$$N(t, x) = C(t) - \int_0^x \rho(t, y)\, dy, \qquad N(0, 0) = 0, \tag{2}$$

where C is total actual incoming cumulative mass, including both merge streams. The active zero-relative-passing condition has saturated congested and adjacent void traces

$$F(\rho_U) - v\rho_U = 0, \qquad \rho_U = \frac{w\kappa}{w + v}, \quad q_U = \frac{w\kappa v}{w + v}, \quad \rho_D = q_D = 0. \tag{3}$$

These are traces on the moving path. They are not an instantaneous universal receiving cap at $x = 0$.

In the congested triangular branch, all changes propagate at speed $-w$. A state emitted from $(\tau, \phi(\tau))$ reaches the entrance at

$$t = T(\tau) = \tau + \frac{\phi(\tau)}{w}, \qquad T'(\tau) = 1 + \frac{v(\tau)}{w} \geq 1. \tag{4}$$

Thus $\tau(t) = T^{-1}(t)$ is unique. In the saturated, isolated subproblem the cumulative receiving envelope and its almost-everywhere derivative are

$$B(t) = \kappa\phi(\tau(t)), \qquad B'(t) = \frac{w\kappa v(\tau(t))}{w + v(\tau(t))} \leq Q. \tag{5}$$

The count formula also follows from a backward wave whose count cost is $w\kappa(t - \tau) = \kappa\phi(\tau)$ and the path's zero count anchor. It agrees with the following-wave construction under the stated saturated data. It is not a theorem for arbitrary sourced or multiple-path roads.

2. A delayed speed change gives a concrete receiving check

Use diagnostic values $u = 30$, $w = 5$, $\kappa = 3/20$ and a prescribed path:

$$v(\tau) = \begin{cases} 5, & 0 \le \tau < 10, \\ 15, & \tau \ge 10, \end{cases} \qquad \phi(\tau) = \begin{cases} 5\tau, & \tau \le 10, \\ 15\tau - 100, & \tau \ge 10. \end{cases} \tag{6}$$

The change is emitted at position 50 m, so arrives at the entrance at $10 + 50/5 = 20$ s. Consequently

$$\tau(t) = \begin{cases} t/2, & t \le 20, \\ (t+20)/4, & t \ge 20, \end{cases} \qquad B(t) = \begin{cases} 3t/8, & t \le 20, \\ 9t/16 - 15/4, & t \ge 20. \end{cases} \tag{7}$$

At $t = 15$ s the current path speed is 15 m/s, but $\tau = 15/2$ s and the entrance still has saturated receiving 3/8 vehicle equivalents/s. Using current speed instead would give 9/16. Integrated to $t = 15$, that shortcut yields

$$B_{\text{shortcut}}(15) = \frac{3}{8}\,10 + \frac{9}{16}\,5 = \frac{105}{16}, \qquad B(15) = \frac{45}{8}, \qquad B_{\text{shortcut}}(15) - B(15) = \frac{15}{16}. \tag{8}$$

The prescribed jump illustrates transport delay. It is not the journal's sampled acceleration output, nor a pair of reached ramp-meter controls. For any admissible sampled piecewise-linear prescribed path, the same emission/arrival construction applies interval by interval.

3. Unused receiving changes actual admission

Let $d(t) \ge 0$ be a prescribed total attempted boundary rate in this diagnostic, with donors plentiful over the finite interval. Put $\bar{d}(t) = \min\{d(t), Q\}$ and $A(t) = \int_0^t \bar{d}(s)\,ds$. The maximal work-conserving admission below the cumulative envelope is

$$L(t) = \max\left\{0, \sup_{0 \le s \le t} [A(s) - B(s)]\right\}, \qquad C(t) = A(t) - L(t) = \inf_{0 \le s \le t} \{B(s) + A(t) - A(s)\}. \tag{9}$$

The equivalence uses $A(0) = B(0) = 0$. L is cumulative held-back mass, not lost population. $0 \le C \le B$; C is nondecreasing and $C' \le Q$. Away from switching points,

$$C'(t) = \begin{cases} \bar{d}(t), & C(t) < B(t), \\ \min\{\bar{d}(t), B'(t)\}, & C(t) = B(t). \end{cases} \tag{10}$$

This is the boundary solution for this isolated road's cumulative receiving constraint, not a proposed universal replacement of native multilane KW.

For constant $v = 5$ m/s, $B(t) = 3t/8$ and $Q = 9/14$. Leave the entrance unused until $t = 10$, then request at least Q:

$$C(t) = \begin{cases} 0, & t \le 10, \\ Q(t-10), & 10 \le t \le 24, \\ 3t/8, & t \ge 24. \end{cases} \qquad \frac{9}{14}(24 - 10) = \frac{3}{8}\,24 = 9. \tag{11}$$

At second 12, actual admission rate is $9/14 > 3/8$: available space absorbs the flow before the induced congestion wave returns. A permanent trace-flow cap would therefore also be wrong, in the opposite direction.

4. The approved priority uses one actual receiving resource

Write R for the admissible instantaneous receiving budget supplied by the compatible field: in this diagnostic $R = Q$ while $C < B$, and $R = B'$ while $C = B$. Actual admission is $C' = \min\{d, R\}$; R is not always B'. With ramp/non-ramp requests $d_r, d_m \geq 0$ and fixed ramp share $p \in [0, 1]$, if $d_r + d_m \leq R$, serve both. Otherwise the already approved N.A scarcity rule is

$$f_r = \min\{d_r, \max\{pR, R - d_m\}\}, \qquad f_m = R - f_r. \tag{12}$$

It satisfies $0 \leq f_r \leq d_r$, $0 \leq f_m \leq d_m$ and $f_r + f_m = R$. Subdivide f_m by compatible non-ramp demand proportions. For $R = 3/8$ and $p = 1/3$, saturated demands receive $f_r = 1/8$ and $f_m = 1/4$. If the ramp asks only $1/20$ and mainline demand is sufficient, the awards are $1/20$ and $13/40$. Both use exactly $3/8$.

For actual transfer rates f_e, donor and recipient changes are paired:

$$\dot{M}_{\text{donor,e}} = -f_e, \qquad \dot{M}_{\text{recipient,e}} = f_e, \qquad \dot{M}_{\text{all}} = a_{\text{offered}} - e_{\text{exits}}. \tag{13}$$

E1 creates descriptors after completed transfers and adds no mass. Its subsequent reset is outside the isolated diagnostic; omitting its future effects does not qualify a repeatedly iterated full task.

5. Own motion, clearing and TST are still separate obligations

The selected source constrained-motion update has

$$v^k = \min\{v^k_{\text{des}}, \widehat{v}^k_{\text{dwn}}\}, \quad \phi^{k+1} = \phi^k + v^k \Delta t, \quad v^{k+1}_{\text{des}} = v^k + a(v^k, \phi^k)\Delta t. \tag{14}$$

The supplied journal identifies $\widehat{v}^k_{\text{dwn}}$ as a numerical estimate over the *ensuing* interval and cites Daganzo/Laval (2005b). The complete adapted own-field/node/source recipe for that estimate has not been transcribed. The fixed-priority operator does not supply it. Nor does exact retention select it automatically.

A conditional local release bound illustrates the clearance distinction. On a level road with free traffic ahead, $a(v) = a_0(1 - v/v_{\max})$, $v_{\max} \geq u$, $a_0 > 0$, initial $v^0 = 0$ and $0 < a_0 \Delta t / v_{\max} \leq 1$, subsequent free-motion sampled speeds are nondecreasing, and after the first update are at least $v_* = \min\{u, a_0 \Delta t\} > 0$. A finite segment of length L_{road} then has the sufficient bound

$$t_{\text{path exit}} \leq \Delta t + \frac{L_{\text{road}}}{v_*}. \tag{15}$$

These are diagnostic sufficient conditions, not newly adopted task bounds. Free traffic ahead is the crucial condition; the bound does not prove clearance of an interacting lane/merge network.

With shared cumulative offered population $A_o(t)$ and initial mass M_0, conservative full accounting gives

$$M(t) = M_0 + A_o(t) - E(t), \quad J_{\text{TST}} = \int_0^\infty M(t)\, dt, \quad J^{(1)}_{\text{TST}} - J^{(2)}_{\text{TST}} = \int_0^\infty [E^{(2)}(t) - E^{(1)}(t)]\, dt. \tag{16}$$

The last identity requires finite objectives or a justified difference limit; normalization by shared offered mass does not change the comparison. Equal complete exits imply equal TST. No nontrivial metering-to-exit witness or full clearing/tail result follows from the local receiving examples.

Disposition. ACT.A preserves the idealized rate task. Local receiving and conditional allocation/accounting are qualified. Full own-motion, accepted-source and simultaneous-restriction coupling remain unbound. At the source/input checkpoint, verify the cited full journal recipe first; an original same-family adaptation must be declared if that input remains unavailable. No physical pulse rule, record approximation, perturbation, replacement solver or numerical traffic experiment is adopted.